



Inspiring Excellence

**CSE370 : Database Systems**  
**Project Report**  
**Project Title : Pawfect Match**

| Group No : _5_, CSE370 Lab Section : _5_, Spring 2026 |                     |                                                                                          |
|-------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------|
| ID                                                    | Name                | Contribution                                                                             |
| 23201544                                              | Tasmin Akter Ridita | EER Diagram,<br>Schema Diagram,<br>Database Creation, 1<br>Feature, 1 Unique<br>Feature. |
| 24101089                                              | Rafia Anan          | EER Diagram,<br>Schema Diagram,<br>Database Creation<br>1 Feature, 1 Unique<br>Feature.  |
| 23101316                                              | Sakib Hossain       | EER Diagram,<br>Schema Diagram,<br>Database Creation,<br>2 Features.                     |

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## **Introduction**

Pet Adoption Management System is a web-based platform designed to make the pet adoption process easier, more organized and more accessible for everyone involved. The idea behind this project is to create a digital space where shelters can manage their animals, potential adopters can find their perfect pet and the entire adoption process can be tracked from start to finish in one place.

In present days, many animal shelters handle adoptions manually through phone calls, paper forms and everything here is offline work. The process is slow and most likely disorganized than the online system.

The platform has several features. It maintains pet profiles by including each animal's name, species, breed, age, temperament and current adoption status. It tracks the health records of every pet which includes vaccinations and medical treatments. It manages a foster parent registry that needs temporary homes before being permanently adopted. It also provides a complete adoption application system where users can apply online and track whether their adoption is pending, approved or denied.

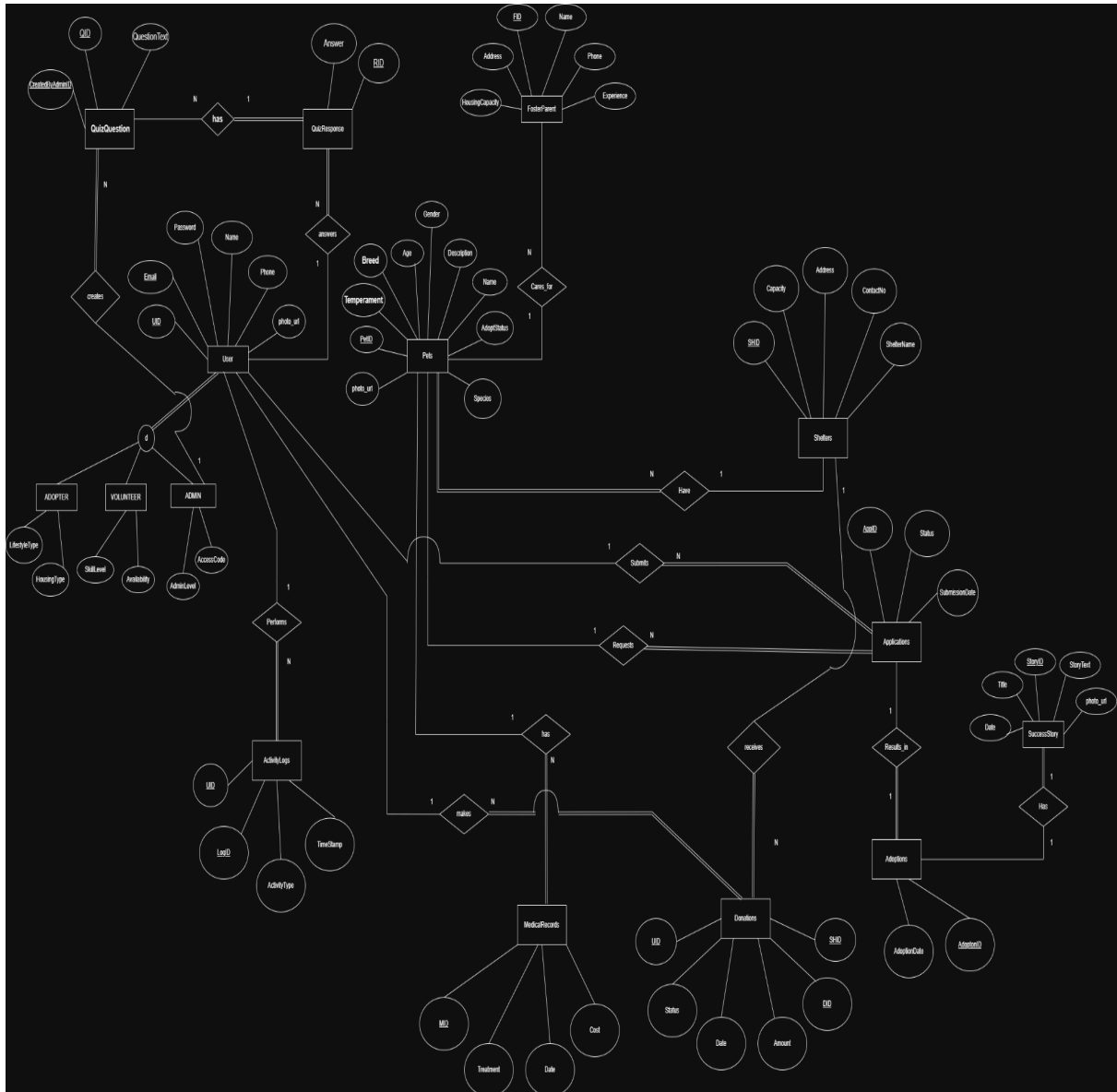
Besides, the system includes two unique additions. The first is a Pet Suggestion Quiz where users answer a few simple questions related to their lifestyle and the system recommends the most suitable pets for them. The second is Success Stories Gallery where families who have adopted can share their experiences and photos which creates a positive and inspiring community around the platform.

The system is using PHP Laravel for the backend, MySQL for the database and a clean responsive frontend for the interface. Every feature is accessible through a structured API which makes the platform reliable and easy to expand in the future.

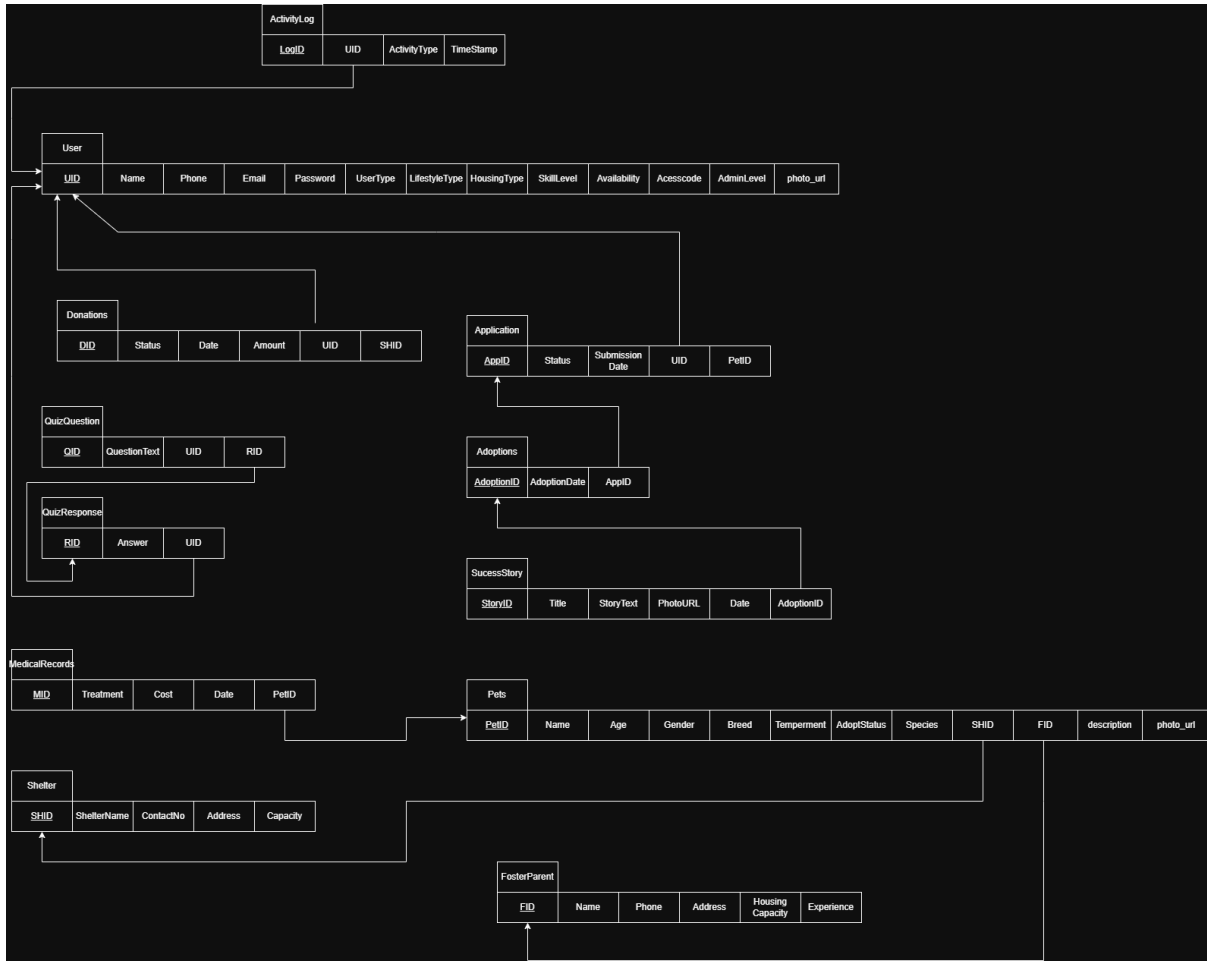
### Project Features

| ID,<br>Name                                  | Features [3 per member] |                                |
|----------------------------------------------|-------------------------|--------------------------------|
| <b>23201544,<br/>Tasmin Akter<br/>Ridita</b> | Ft 1                    | Pet Profile Management         |
|                                              | Ft 2                    | Success Stories Gallery        |
|                                              | Ft 3                    |                                |
| <b>24101089,<br/>Rafia Anan</b>              | Ft 1                    | Adoption Application System    |
|                                              | Ft 2                    | Quiz System for Pet Suggestion |
|                                              | Ft 3                    |                                |
| <b>23101316,<br/>Md Sakib<br/>Hossain</b>    | Ft 1                    | Health Record Management       |
|                                              | Ft 2                    | Foster Parent Registry         |
|                                              | Ft 3                    |                                |

# ER/EER Diagram



# Schema Diagram



## Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

- a. Our converted Schema is in 1NF because it does not contain any composite or multivalued attributes or any nested relations.
- b. Our converted Schema is in 2NF because it is in 1NF and does not contain any partial functional dependencies.
- c. Our converted Schema is in 1NF because it is in 2NF and does not contain transitive dependencies.

## **Frontend Development**

**Contribution of ID : 23201544, Name : Tasmin Akter Ridita**

Dashboard

**Contribution of ID : 24101089, Name : Rafia Anan**

Home Page Layout

**Contribution of ID : 23101316, Name : Md Sakib Hossain**

Ui and Ux

The frontend of the Pet Adoption Management System was developed using React. It provides a user-friendly interface for browsing pets, viewing pet details, submitting adoption applications, taking quizzes, reading success stories, logging in/registering, and accessing the admin dashboard.



```
C:\Users\De11 7480\pet-adoption-management-system\frontend>npm run dev
```

```
> pet-adoption-frontend@1.0.0 dev
> vite
```

```
VITE v6.4.2 ready in 1238 ms
```

```
Local: http://localhost:5173/
```

```
Network: use --host to expose
```

```
press h + enter to show help
```

localhost:5173/dashboard

english to bangla... Dashboard | buX docs.google.com SamOnline BRAC University S... Progress Perplexity Coursera | Online... Admin

**Pawfect Match**  
Adopt love, one paw at a time

Home Browse Pets Take the Quiz Success Stories **Dashboard** Pawfect Admin

**TOTAL PETS**

2

**AVAILABLE PETS**

0

**PENDING APPLICATIONS**

1

**TOTAL ADOPTIONS**

1

**Manage Pets**

- Manage Users
- Success Stories
- Manage Shelters
- Foster Parents
- Applications
- Health Records
- Activity Logs

**CURRENT SECTION**

**Manage Pets** Refresh Add New

| PET ID | PHOTO URL                                                                           | NAME    | SPECIES | BREED            | AGE | GENDER | ADOPT STATUS | ACTIONS     |
|--------|-------------------------------------------------------------------------------------|---------|---------|------------------|-----|--------|--------------|-------------|
| 4      |  | pumpkin | Dog     | Golden Retriever | 3   | Male   | Adopted      | Edit Delete |
| 3      |  | Tom5    | Cat     | Persian          | 6   | Male   | Pending      | Edit Delete |

## **Backend Development**

**Contribution of ID : 23201544, Name : Tasmin Akter Ridita**

Pet, Success Story, Donation

**Contribution of ID : 24101089, Name : Rafia Anan**

Adoption, Application, Quiz Question, Quiz Response

**Contribution of ID : 23101316, Name : Md Sakib Hossain**

User, Shelter, Medical Records, Foster Parent

**The backend was developed using Laravel and connected with a MySQL database. It handles user registration, login, admin access, pet management, adoption applications, success stories, quiz responses, image uploads, and API communication with the React frontend.**

```
< > 127.0.0.1:8000/api/pets
english to bangla... Dashboard | buX docs.google.com SamOnline BRAC University S... Progress Perplexity Coursera | Online... Admin >>
Pretty-print
[{"pet_id":4,"name":"pumpkin","age":3,"gender":"Male","breed":"Golden Retriever","temperament":"Playfull","adopt_status":"Adopted","species":"Dog","shid":8,"fid":null,"description":"its a very playful dog. blaiaaaaaaaaaaaaaaaaaaaaaa","photo_url":"\\storage\\pets\\hFiEtXc9XsdmFobntoLmm6fTmEU1P1Sycgyobnav.jpg"}, {"pet_id":3,"name":"Tom5","age":6,"gender":"Male","breed":"Persian","temperament":"Playfull","adopt_status":"Pending","species":"Cat","shid":8,"fid":1,"description":"No description has been added for this pet yet.","photo_url":"\\storage\\pets\\K6jC2vM1St15fy13KfbbXzLUCdNJS0tQM1sZZ4H2.jpg"}]
```

```
C:\Users\Dell 7480\pet-adoption-management-system\backend>php artisan serve --host=127.0.0.1 --port=8000

INFO Server running on [http://127.0.0.1:8000].

Press Ctrl+C to stop the server

2026-05-03 14:24:59 /api/pets ..... ~ 1s
2026-05-03 14:25:00 /api/shelters ..... ~ 0.65ms
2026-05-03 14:25:00 /api/foster-parents ..... ~ 507.27ms
2026-05-03 14:25:00 /api/medical-records ..... ~ 505.98ms
2026-05-03 14:25:00 /api/users ..... ~ 512.79ms
2026-05-03 14:25:01 /api/applications ..... ~ 519.34ms
2026-05-03 14:25:01 /api/success-stories ..... ~ 5.13ms
2026-05-03 14:25:01 /api/activity-logs ..... ~ 514.15ms
2026-05-03 14:25:01 /api/adoptions ..... ~ 511.60ms
2026-05-03 14:25:01 /api/me ..... ~ 509.70ms
2026-05-03 14:25:02 /api/pets ..... ~ 509.28ms
2026-05-03 14:25:02 /api/shelters ..... ~ 3.33ms
2026-05-03 14:25:02 /api/foster-parents ..... ~ 504.13ms
2026-05-03 14:25:02 /api/medical-records ..... ~ 504.52ms
2026-05-03 14:25:03 /api/users ..... ~ 510.68ms
2026-05-03 14:25:03 /api/applications ..... ~ 511.98ms
2026-05-03 14:25:03 /api/success-stories ..... ~ 508.80ms
2026-05-03 14:25:03 /api/activity-logs ..... ~ 508.31ms
2026-05-03 14:25:03 /api/adoptions ..... ~ 507.77ms
2026-05-03 14:25:04 /api/me ..... ~ 507.12ms
2026-05-03 14:25:04 /api/pets ..... ~ 3s
2026-05-03 14:25:04 /api/shelters ..... ~ 3s
2026-05-03 14:25:08 /api/foster-parents ..... ~ 508.47ms
```

## **Source Code Repository**

**Github Repository :**

**<https://github.com/sakibhossain0/pet-adoption-management-system.git>**

## **Conclusion**

This project was a good learning experience for all of us. We got to apply what we studied in class into something that actually works and solves a real problem. From designing the database to building the API and planning the frontend, every part of the process taught us something valuable.

Throughout the development process, we applied the concepts we learned in our coursework, from database design and normalization to backend API development and frontend interface planning. The ER diagram, schema design, and normalized tables formed a strong foundation that made the development phase more organized and efficient. The Laravel backend provides a clean and reliable way to manage data through structured API endpoints, while the frontend design ensures that the system is accessible and easy to use for anyone regardless of technical background.

Each feature of the system, for instance pet profile management, health record tracking, foster parent registry, adoption applications, the quiz system, and the success stories gallery was carefully planned to address a specific gap in how pet adoptions are currently handled. Together they form a complete solution rather than a collection of isolated tools.

This project also strengthened our ability to work collaboratively, divide responsibilities and integrate different parts of a system into one cohesive product. The challenges we faced during development gave us practical experience that goes beyond what any textbook can provide.

## References

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